

High Temperature Limit of the Confining Phase

Source-backed arXiv sample

Abstract

The deconfining transition in non-Abelian gauge theory is known to occur by a condensation of Wilson lines. By expanding around an appropriate Wilson line background, it is possible at large N to analytically continue the confining phase to arbitrarily high temperatures, reaching a weak coupling confinement regime. This is used to study the high temperature partition function of an $SU(N)$ electric flux tube. It is found that the partition function corresponds to that of a string theory with a number of world-sheet fields that diverges at short distance.

1 Source Metadata

This sample is based on arXiv record `hep-th/9109007`. Authors: Joseph Polchinski.

2 Extracted Prose 1

The two-point function vanishes at infinity, so the expectation values of the order parameters are zero and the $U(1)$ symmetry is unbroken. The Wilson line W is an external quark source, while the W^a correspond to combinations of sources in higher representations. The T_a are thus the effective string tensions for flux in the various representations. It is very useful to consider also a dual point of view, in which W is regarded as a spatial coordinate. Then

3 Extracted Prose 2

are the masses of winding states, states with electric flux in the periodic direction. At sufficiently small g these winding states become tachyonic (presumably the singly-wound W becomes negative first) and the theory makes a transition to a phase in which the order parameters have expectation values and the $U(1)$ symmetry is broken. W These expectation values mean that an isolated quark source has finite energy: the theory no longer confines. In the bosonic string theory there are also winding states, solitons with vertex operators V and masses